PhD Math Camp

Relations and Functions

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Relations

Understanding Relations

- **Definition:** A relation is a set of ordered pairs, a subset of the Cartesian product of two sets.
- **Components:** Consists of a domain and a range.
- **Visualization:** Often represented using Venn diagrams or graphs.
- **Non-Numeric Example:** Consider the relation R on the set $X = \{\text{students}\}$ and $Y = \{\text{majors}\}$:
  - Let $X = \{\text{Jordan}, \text{Bob}, \text{Carol}\}$
  - Let $Y = \{\text{History}, \text{Engineering}, \text{Biology}\}$
  - Define R as follows:

    $$R = \{(\text{Jordan}, \text{History}), (\text{Bob}, \text{Engineering}), (\text{Carol}, \text{Biology})\}$$

  - Here, R relates each student to their major.

Basic Types of Relations

- **Reflexive:** Every element is related to itself.
- **Symmetric:** If x is related to y, then y is related to x.
- **Transitive:** If x is related to y and y is related to z, then x is related to z.

Special Types of Relations

- **Functions:** Each element of the domain is associated with exactly one element of the range.
- **Equivalence Relations:** Reflexive, symmetric, and transitive.
- **Order Relations:** Reflexive, antisymmetric, and transitive. Important in hierarchical structures.

Ex. (Applications in social science)

- **Social Networks:** Analysis of friendships, social media connections.
- **Organizational Structures:** Supervision and hierarchy in corporations.

Toy Example: Three Relations on the Same People

Take four students, each with a major and a year in the program:

Student	**Major**	**Year**
Jordan	History	3
Bob	Engineering	2
Carol	Biology	1
Dana	History	4

On the set $P = \{\text{Jordan}, \text{Bob}, \text{Carol}, \text{Dana}\}$ define three relations:

- S: **has the same major as**
- A: **is at least as advanced as** (year $\geq$ year)
- F: **is a friend of**, where the friendships are Jordan–Bob and Bob–Carol

Toy Example: Which Properties Hold?

		Reflexive	Symmetric	Antisym.	Transitive
S	same major	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
A	at least as advanced	$\checkmark$	$\times$	$\checkmark$	$\checkmark$
F	is a friend of	$\times$	$\checkmark$	$\times$	$\times$

- S is reflexive, symmetric and transitive, so it is an **equivalence relation**. It partitions the students into classes: $\{\text{Jordan}, \text{Dana}\}$, $\{\text{Bob}\}$, $\{\text{Carol}\}$.
- A is reflexive, antisymmetric and transitive, so it is an **order relation**. It ranks them: Carol, Bob, Jordan, Dana.
- F is neither: Jordan is a friend of Bob and Bob of Carol, but Jordan and Carol are not friends, so F fails transitivity.

Functions

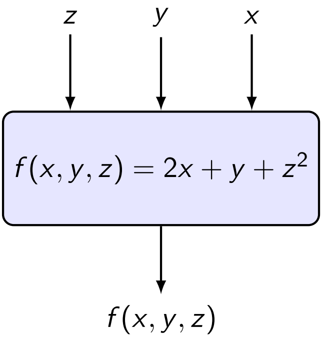
Introduction of Functions

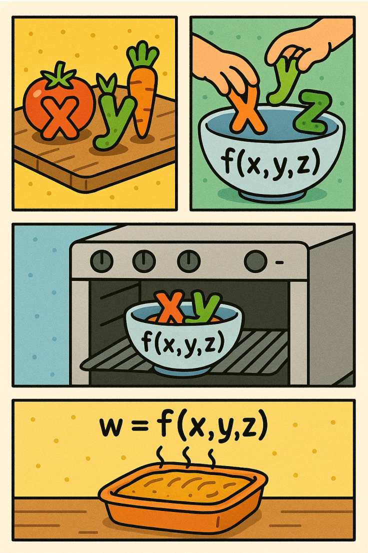
Def. A function is a mapping from one defined space to another.

$$f : A \to B$$

- A is the domain of the function, B is the codomain of the function.
- The set of all values actually reached by running each $x \in A$ through f is known as the range (or image).

Function as a Box, Process, or Recipe





Introduction of Functions

Functions can be "many-to-one" or "one-to-one", but *cannot* be "one-to-many".

Examples:

- **Function, many-to-one:** $y = f(x) = x^2$
- **Function, one-to-one:** $y = f(x) = 2x + 4$
- **Relation that is not a function:** $y^2 = 5x$

Relation that is not a function

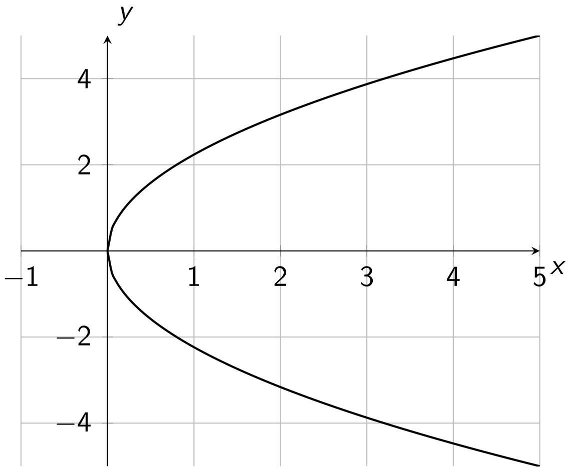


Figure: The parabola described by $y^2 = 5x$

Linear Functions and Slope

Def. (Linear function) A function of the form

$$f(x) = b_0 + b_1 x,$$

whose graph is a straight line. In the familiar form $y = mx + b$, the slope is $m = b_1$ and the intercept is $b = b_0$.

Def. (Slope) The rate of change of f, the same everywhere on the line:

$$b_1 = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Read b_1 as: a one-unit increase in x changes $f(x)$ by b_1, *no matter where you start*. That constancy is exactly what makes the function linear.

Why Slope Is the Point: A Regression

Ex. (Political participation) Let Y be political participation, X_1 **affective polarization** (the feeling-thermometer gap between one's own party and the out-party), and X_2 **internal efficacy** (belief that one can understand and influence politics):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon.$$

- β_1 is a **slope**: the change in expected participation for a one-point widening of the thermometer gap, holding efficacy fixed.
- β_0 is the **intercept**: expected participation for someone who rates both parties equally and reports no efficacy.
- ε collects everything the two predictors do not account for.

Dimensions

Dimensions

$$f : \mathbb{R}^1 \to \mathbb{R}^1$$

Examples:

- $f(x) = x + 1$
- $f(x) = 2x + 3$
- $f(x) = x^2 - 4x + 4$
- $f(x) = \sin(x)$
- $f(x) = e^x$

Dimensions

$$f : \mathbb{R}^2 \to \mathbb{R}^1, \quad \text{e.g. } f(x,y) = x^2 + y^2$$

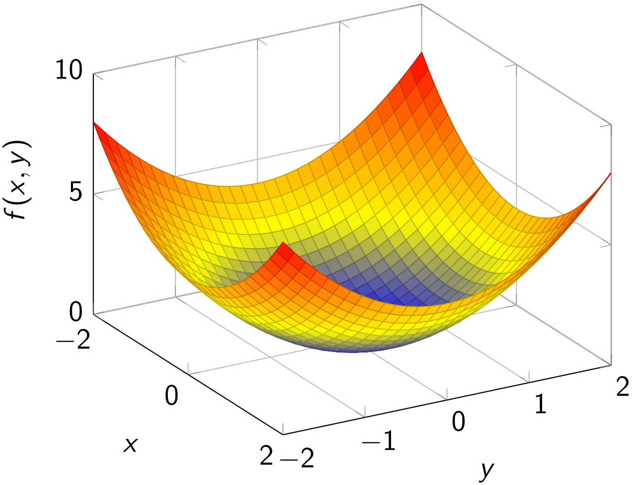
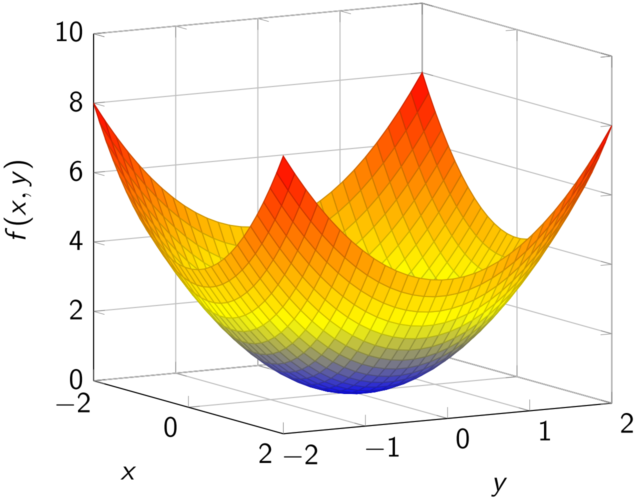


Figure: $f(x,y) = x^2 + y^2$

Dimensions

$$f(x, y) = x^2 + y^2$$



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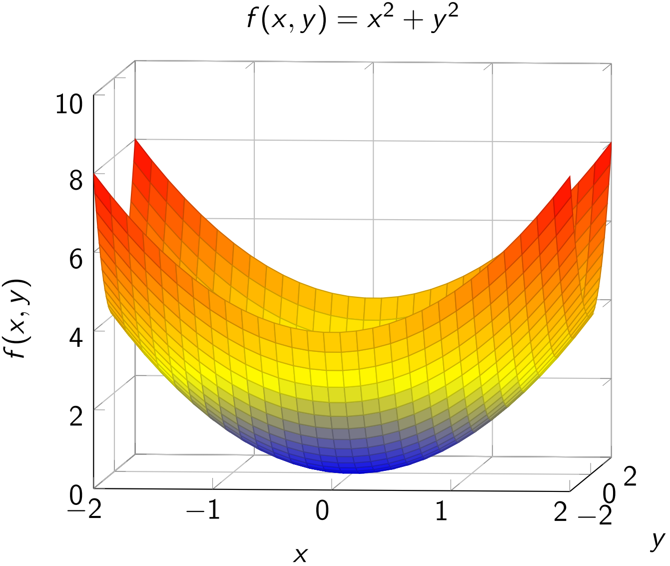


Dimensions

$$f : \mathbb{R}^2 \to \mathbb{R}^1, \quad \text{e.g. } f(x, y) = x^3 + 2y^2$$

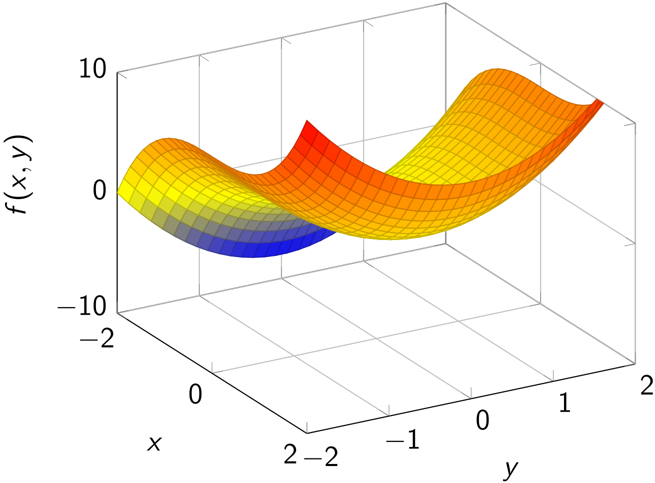
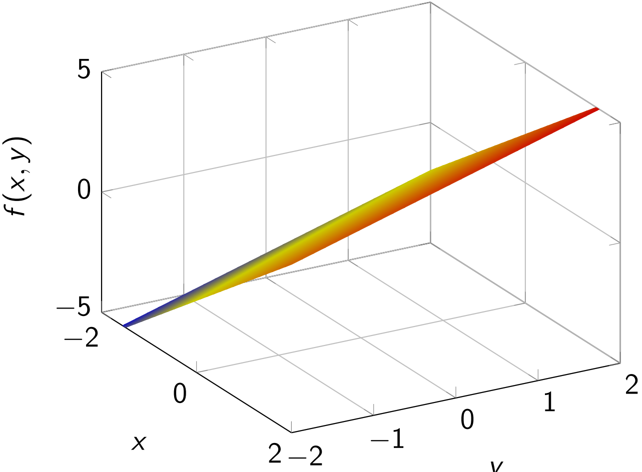


Figure: $f(x, y) = x^3 + 2y^2$

Dimensions

$$f : \mathbb{R}^2 \to \mathbb{R}^1, \quad \text{e.g. } f(x, y) = 2x + y$$

For each ordered pair (called tuple) (x, y), $f(x, y)$ assigns the number $2x + y$



Composition and Inverses

Function Composition (Nested Function)

Def. If we have $f : A \to B$ and $g : B \to C$, then $g(f(x)) : A \to C$.

- Can also be denoted as $g \circ f(x)$.
- E.g., if $f(x) = 2x$ and $g(x) = x^3$, then $g \circ f(x) = (2x)^3 = 8x^3$.

Inverse Function

Def. Function that when composed with the original function returns the identity function.

$$f(x) = y \Leftrightarrow f^{-1}(y) = x$$

i.e. $f^{-1}(f(x)) = x$

- E.g., if $f(x) = 3x + 2$, then $f^{-1}(x) = \frac{x-2}{3}$.
- If $f(x) = x^3 + 2$, then $f^{-1}(x) = \sqrt[3]{x-2}$.

Polynomials and Equations

Polynomial Functions

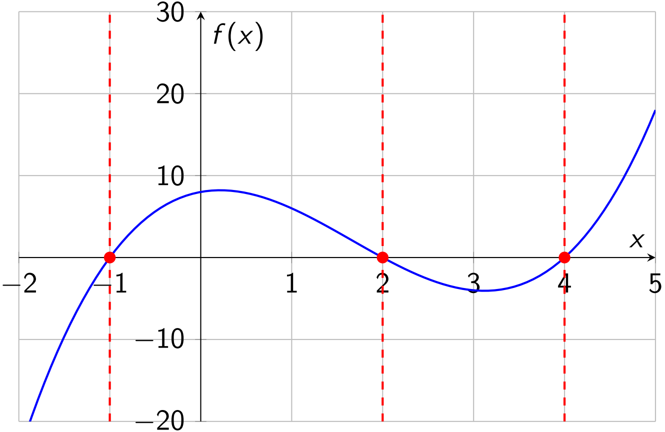
Polynomial functions of x are functions that have components that raise x to some power, e.g.,

$$f(x) = x^3 + x^2 + x + 1$$

Solving Equations

Roots of a polynomial function: where the curve of the function crosses the x-axis, i.e. $f(x) = 0$.

- E.g., $x^3 - 5x^2 + 2x + 8 = 0$
- **Factoring polynomials:**
  $x^3 - 5x^2 + 2x + 8 = (x+1)(x-2)(x-4) = 0$



Solving Quadratic Equations by Formula

Roots of a quadratic polynomial: if $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Exercise:** Solve for x: $f(x) = x^2 + 3x - 4 = 0$

Logarithms and Exponentials

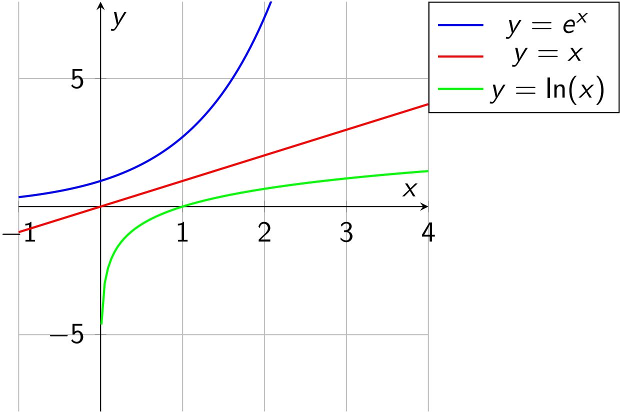
Logarithmic and Exponential

$$y = \log_a(x) \Leftrightarrow a^y = x$$

- The log function can be thought of as an inverse for exponential functions.
- a is referred to as the "base" of the logarithm.

Logarithmic and Exponential with e

A comparison of the functions $y = e^x$, $y = x$, and $y = \ln(x)$ on the same graph.



Properties of Exponential Functions

- $a^x \cdot a^y = a^{x+y}$
- $a^{-x} = \frac{1}{a^x}$
- $\frac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{xy}$
- $a^0 = 1$

Basic Properties of Logarithms

- **Zero/One** $\log_b(1) = 0$
- **Multiplication** $\log(x \cdot y) = \log(x) + \log(y)$
- **Division** $\log\left(\frac{x}{y}\right) = \log(x) - \log(y)$
- **Exponentiation** $\log(x^y) = y\log(x)$
- **Basis** $\log_b(b^x) = x$, and $b^{\log_b(x)} = x$

These properties hold for any base.

- **Exercise:** Simplify $\log\left(\frac{x^9 y^5}{z^3}\right)$

Monotonic Functions

Monotonic Functions

Def. A function $f(x)$ is said to be monotonic if it preserves order — that is, if it is either entirely non-increasing or non-decreasing. It is **strictly** monotonic when the inequality is strict:

$$x_1 < x_2 \implies f(x_1) < f(x_2).$$

- For example, $f(x) = 2x + 3$ is strictly increasing.
- $\log(x)$ is strictly increasing on $(0, \infty)$.
- $f(x) = x^2$ is *not* monotonic on $\mathbb{R}$: it decreases, then increases.

Monotonic Functions (cont'd): Why This Matters

Prop. If g is strictly increasing, then f and $g(f)$ are maximized in the same place:

$$\arg\max_x f(x) \;=\; \arg\max_x g\big(f(x)\big).$$

Such a transformation changes the *height* of the maximum but not its *location* — so a hard objective may be swapped for an easier one with the same maximizer.

Ex. (Maximum likelihood) The likelihood is a product, awkward to differentiate: $L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$. Since log is strictly increasing, maximize the log-likelihood instead — a sum: $\ell(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta)$. Both peak at the same $\hat{\theta}$.